

Exercise 1: Mocking and Stubbing

ExternalApi.java:

```
package com.exercise1;

public interface ExternalApi {

    String getData();

}
```

MyService.java:

```
package com.exercise1;

public class MyService {

    private final ExternalApi api;

    public MyService(ExternalApi api) {

        this.api = api;

    }

    public String fetchData() {

        return api.getData();

    }

}
```

MyServicesTest.java:

```
package com.exercise1;
```

```
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.assertEquals;

import static org.mockito.Mockito.*;

public class MyServiceTest {

    @Test
    void testExternalApi() {

        // Arrange

        ExternalApi mockApi = mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");

        MyService service = new MyService(mockApi);

        // Act

        String result = service.fetchData();

        // Assert

        assertEquals("Mock Data", result);
    }
}
```

Output:

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.exercise1.MyServiceTest
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe:objectFieldOffset has been called by net.bytebuddy.dynamic.loading.ClassInjector$UsingUnsafe$Dispatcher$CreationAction (file
:/C:/Users/adity/.m2/repository/net/bytebuddy/byte-buddy/1.14.9/byte-buddy-1.14.9.jar)
WARNING: Please consider reporting this to the maintainers of class net.bytebuddy.dynamic.loading.ClassInjector$UsingUnsafe$Dispatcher$CreationAction
WARNING: sun.misc.Unsafe:objectFieldOffset will be removed in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.610 s -- in com.exercise1.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.398 s
[INFO] Finished at: 2026-06-21T21:28:21+05:30
[INFO] -----
```